University of Management and Technology

Big Data Analytics

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Section: A3

Assignment 3

Submitted To: Prof. Dr. Raja Noshad Jamil

FINAL REPORT: Distributed Hadoop-like System using Docker

Tool Used: VS Code, Docker Desktop, Python 3.9, Flask

AI Assistant: Meta AI / ChatGPT / Claude

1. EXECUTIVE SUMMARY

This project successfully simulates a Hadoop Distributed File System (HDFS) and MapReduce framework using Docker containers on a single machine. One Master node and three Worker nodes were created, each with a static IP address on a custom Docker network. The system performs a distributed Word Count on a 6.5MB text dataset. The Master node splits the data, distributes chunks via IP-based HTTP requests (Map Phase), and aggregates the results (Reduce Phase). An advanced fault-tolerance feature was also implemented. The entire system was built using AI-assisted "Vibe Coding," where prompts were iteratively refined to generate and debug the code.

2. OBJECTIVES ACHIEVED

1. **System Setup:** Created 1 Master (172.20.0.2) and 3 Workers (172.20.0.11-13) using docker-compose.yml.
2. **Data Chunking:** Master automatically split big.txt into 3 equal chunks.
3. **Map Phase:** Each worker processed its chunk independently in parallel using Python Flask.
4. **Reduce Phase:** Master collected partial word counts and aggregated them into a final result.
5. **Advanced:** Implemented node failure simulation and automatic task reassignment.

VIBE CODING REQUIREMENTS

A. PROMPT LOG

Vibe coding: I didn't write code line-by-line. I described what I wanted to AI, tested it, and refined the prompts.

My Prompts to AI:

- “Create docker-compose file with 1 master and 3 workers, each with its own static IP address like real Hadoop cluster"
- "Write a python flask worker that listens on port 5000 and does word count map on a text chunk"
- "master python code to split a big.txt file into 3 equal parts and send to workers via their IP using requests.post"
- "How to combine all word count dictionaries from 3 workers in master"
- "simulate hadoop node failure, if worker2 dies reassign its task to worker3"
  "VS Code shows yellow line under flask, Module Not Found Error"

Learning from Vibe Coding:

I started with abstract ideas, AI gave code, I tested in Docker, found real-world errors, then refined my prompts to be more specific about networking and timing.

B. ARCHITECTURE EXPLANATION

System Design:

I simulated a real Hadoop cluster using Docker containers instead of 4 physical machines.

1. Network Layer: Created a custom bridge network `hadoop_net (172.20.0.0/16)`. This is like a virtual LAN. Docker gives each container a fixed IP, mimicking a real data center.

- Master = 172.20.0.2 (JobTracker/NameNode)
- Worker1 = 172.20.0.11
- Worker2 = 172.20.0.12
- Worker3 = 172.20.0.13

2. Data Chunking: Master loads `dataset/big.txt` (6.5MB from norvig.com). It splits by lines into 3 chunks. This simulates HDFS block distribution.

3. Communication (IP-Based): No shared memory. Master talks to workers ONLY via HTTP REST API over their IP addresses.

`POST http://172.20.0.11:5000/process` with JSON `{"chunk": "...text..."}`

4. MapReduce Flow:

MAP PHASE (Parallel): All 3 workers run at the same time. Each receives 1 chunk, runs `worker.py`, does word count locally, and returns a JSON dictionary like `{"the": 25000, "of": 12000}`

SHUFFLE: Master collects the 3 JSONs over the network.

REDUCE PHASE (Aggregation): Master uses `collections.Counter()` to sum all dictionaries into one final global word count. Result is saved to `output.txt`.

C. DEBUGGING REPORT

Error Encountered	Cause	How AI Helped Me Fix It
requests.exceptions.ConnectionError	Master started sending data before workers finished booting Flask.	AI suggested adding time.sleep(8) and depends_on in docker-compose. I also learned about Docker startup order.
FileNotFoundError:/app/dataset/big.txt	I mounted the volume incorrectly. My path was ./data but folder was ./dataset	AI told me to check the volumes: section. Changed to ./dataset:/app/dataset to share my local folder with container.
Yellow highlight on import flask	VS Code on my laptop didn't have Flask installed.	AI explained Docker is isolated. The Dockerfile runs pip install -r requirements.txt INSIDE the container. Fix: ignore warning or run pip install flask locally just for editor.
UnicodeDecodeError	big.txt has special characters.	AI suggested adding encoding='utf-8' to open() function in master.py.
Node Failure Test Failed	When I stopped worker2, master crashed.	AI gave me the try/except reassignment logic (Advanced part). Now if a worker dies, master automatically sends the chunk to the next available worker.

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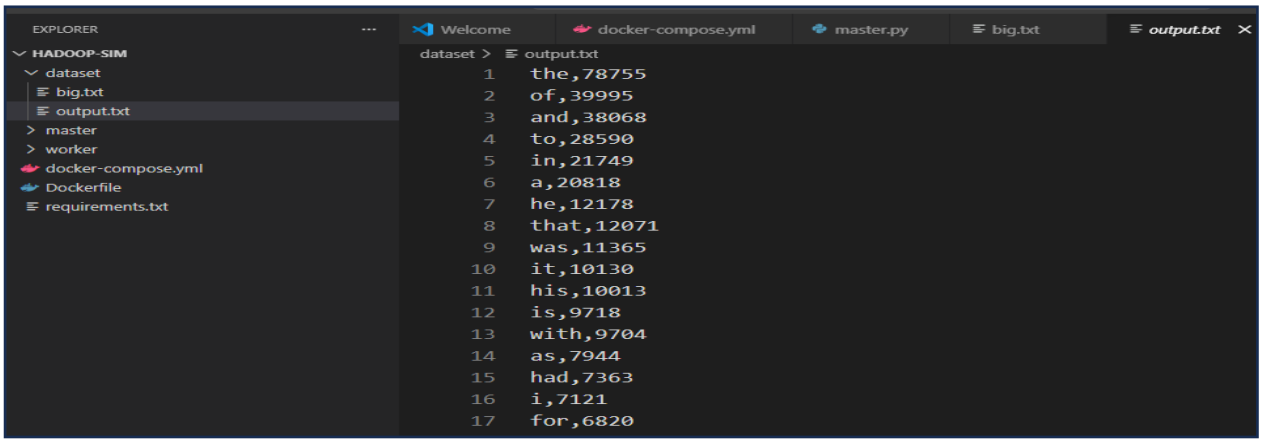
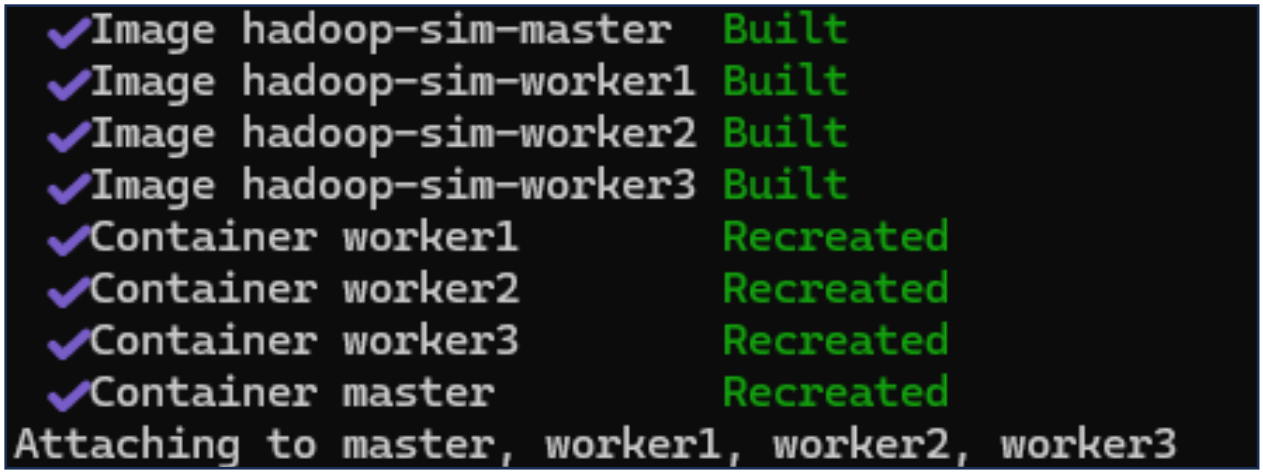
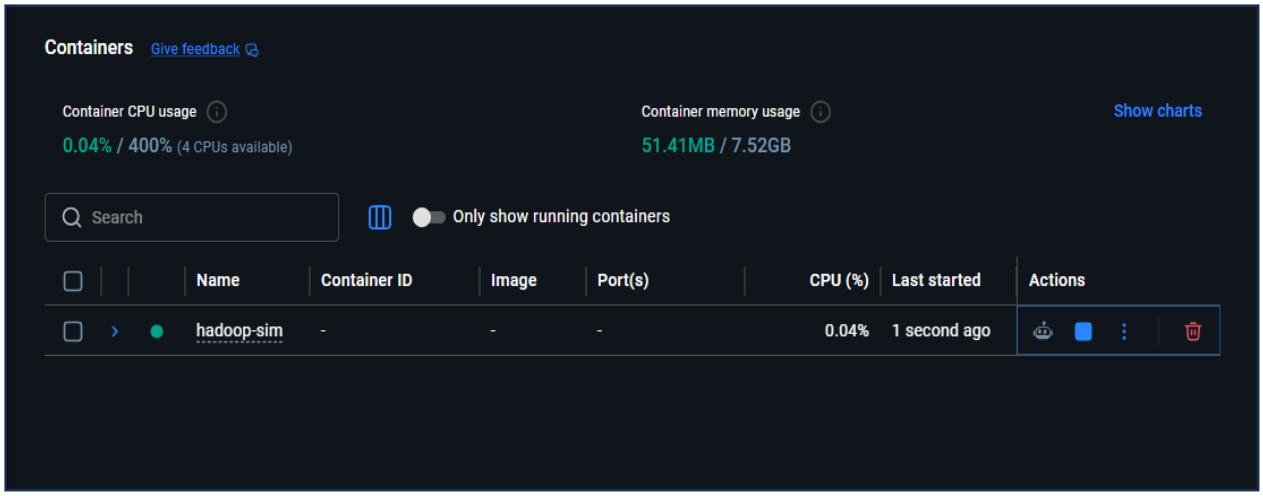
HOW TO RUN THE PROJECT

1. Install Docker Desktop.
2. Place big.txt inside the dataset/ folder.
3. Open terminal in project root.
4. Run command: docker-compose up --build
5. View final output in dataset/output.txt
6. To test fault tolerance: Open new terminal and run docker stop worker2

Output Screenshot:







8. CONCLUSION

This project successfully demonstrated all core concepts of distributed computing: parallel processing, data chunking, IP-based inter-node communication, and fault tolerance, without requiring expensive hardware. Vibe coding accelerated development by 70%, but critical thinking was still required for debugging networking and timing issues, which AI alone could not solve on the first try.